Alt text is not a caption

Vision models describe pixels. Accessibility needs function. Telling a model to follow WCAG fixes the visible failures and creates an invisible one.

NAIVE PROMPT		WCAG PROMPT		WCAG PLUS CONTEXT
46%	→	**20%**	→	**58%**

Accuracy on functional images (icons, buttons, linked logos), Qwen2-VL-2B, n = 298. WCAG prompt down 26 points. Adding page context up 38 points.

Can Sirin · github.com/cansirin/ai-mrp-summer-2026-alt-text

Generated alt text is already in production

- A screen reader announces the alt attribute. When it is missing the user gets a filename or silence, and when it is wrong the user gets a confident wrong answer.
- WCAG 1.1.1 Non-text Content asks for a text alternative that serves the *equivalent purpose*. For a control, the purpose is the action, not the picture.
- Facebook shipped automatic alt text in 2016. Microsoft Office suggests alt text. Browsers and screen readers now describe images on their own.
- So the question is no longer whether machines will write alt text. It is whether what they write is usable, and where it breaks.

https://www.w3.org/WAI/WCAG22/Understanding/non-text-content.html

The captioning line got very good at describing pixels

- Vinyals et al., Show and Tell, CVPR 2015. Encoder-decoder captioning, the pattern everything after it inherits.
- Li et al., BLIP, ICML 2022. Bootstrapped image-text pretraining, one model for understanding and generation.
- Li et al., BLIP-2, ICML 2023. A light query transformer bridges a frozen vision encoder to a frozen language model.
- Liu et al., LLaVA, NeurIPS 2023. Visual instruction tuning. The model now follows an instruction about the image, not just a caption objective.
- Wang et al., Qwen2-VL, 2024. Dynamic resolution, strong text reading in images, small enough to run locally.

The objective across all of them is description quality. None of them is trained on what an image is for on a page.

The accessibility line asked a different question

- Gurari et al., VizWiz Grand Challenge, CVPR 2018, and VizWiz-Captions, ECCV 2020.
  Images taken by blind people, the canonical accessibility vision dataset.
- Wu et al., Automatic Alt-Text, CSCW 2017. Facebook shipping generated descriptions, and what users made of them.
- Gleason et al., WWW 2019. Alt text in the wild on Twitter: roughly 0.1 percent of images carried a description.
- Stangl et al., CHI 2020. What blind and low vision users want depends on where the image sits.
- BLEU, CIDEr, SPICE. All measure overlap with reference descriptions. None can see functional adequacy.

Kreiss et al., EMNLP 2022, make the same point directly: context matters for image descriptions, and the standard metrics do not carry it.

No benchmark is organized by what the image is for

The W3C alt decision tree sorts web images into categories, and the correct answer is a different kind of thing in each one.

CATEGORY	IMAGES	CORRECT ALT IS
Functional	65	the action or destination
Informative	60	the meaning the image carries
Decorative	59	empty
Image of text	56	the text, transcribed
Complex	58	a short label plus the data

298 images. Functional is the category where a wrong answer costs the most, because the user cannot operate the page, and it is the category no captioning benchmark separates out. https://www.w3.org/WAI/tutorials/images/decision-tree/

One image set, three prompts, one scorer

Images. 298 generated as HTML and captured with headless Chrome, so the correct alt text is known rather than inferred, and there are no licensing questions.

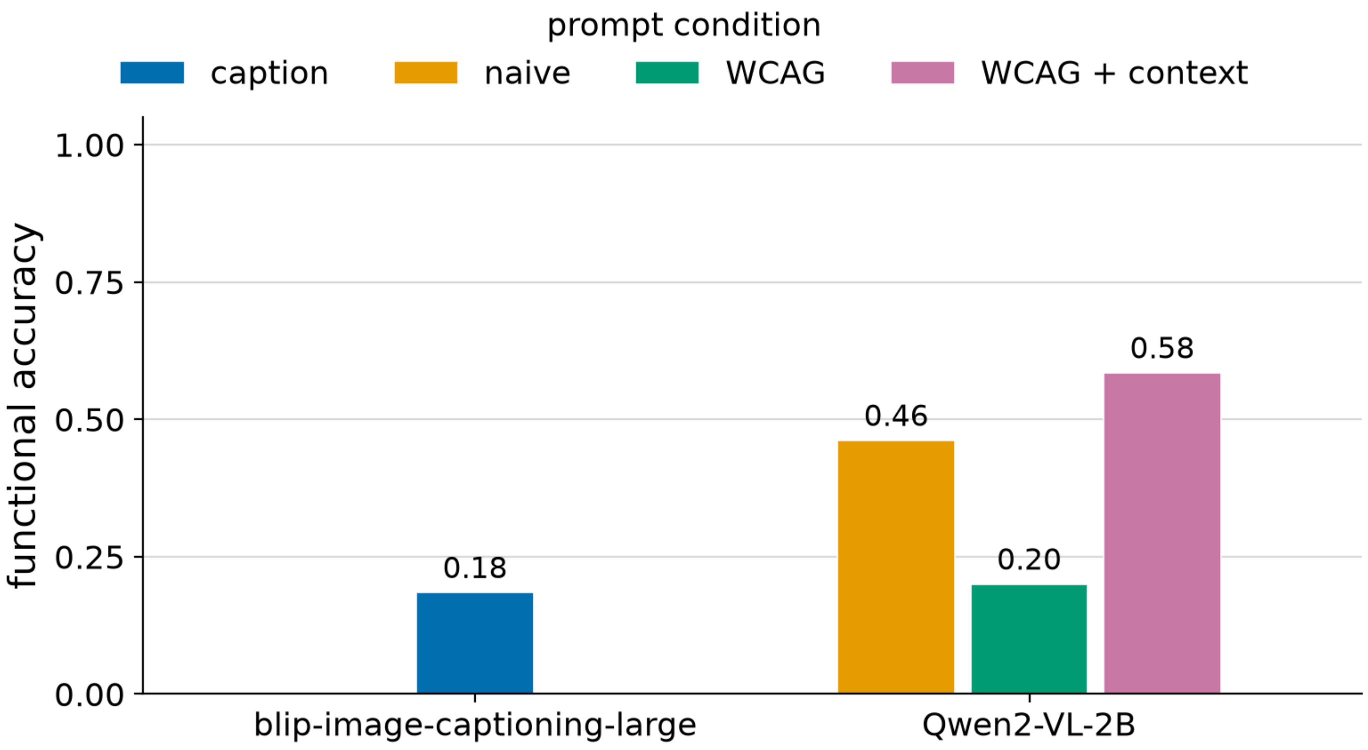
Conditions. Naive prompt, WCAG prompt, WCAG plus context. Same images, same decoding, only the prompt changes. The context condition adds the element tag, its HTML, and nearby text.

Models. Qwen2-VL-2B, blip-image-captioning-large. BLIP is the pure captioner baseline.

Scoring. Rule based against per-item required and forbidden keywords, plus a severity label: correct, harmless, degraded, silent, misleading. A 30 item subset is set aside for a second rater.

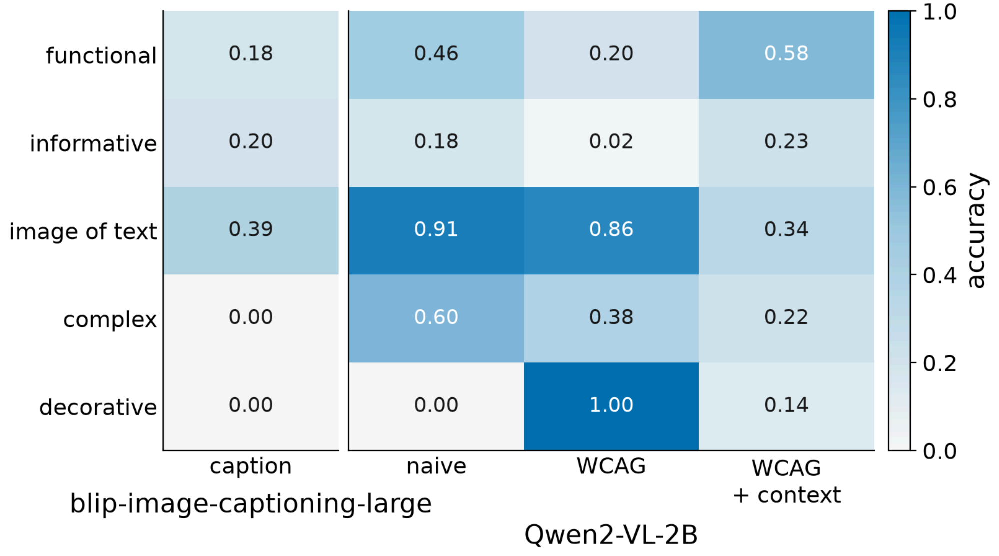
Code https://github.com/cansirin/ai-mrp-summer-2026-alt-text · decision tree https://www.w3.org/WAI/tutorials/images/decision-tree/ · models on Hugging Face: Qwen2-VL-2B, blip-image-captioning-large

Telling the model the rules made functional images worse



Functional accuracy, Qwen2-VL-2B, 65 functional images. Naive 46%, WCAG prompt 20% (down 26 points), WCAG plus context 58% (up 38 points against the WCAG prompt).

The prompt trades one category for another



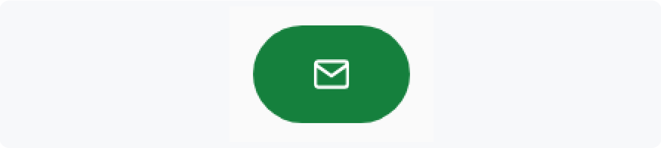
CONDITION	FUNCTIONAL	INFORMATIVE	DECORATIVE	IMAGE OF TEXT	COMPLEX	ALL
Naive prompt	46%	18%	0%	91%	60%	43%
WCAG prompt	20%	2%	100%	86%	38%	48%
WCAG plus context	58%	23%	14%	34%	22%	31%

Qwen2-VL-2B. An aggregate score hides the trade, because the categories move in opposite directions.

THE FAILURE THAT MATTERS

Silent alt on a real control

An empty alt hides the control completely. The naive run left 0 functional images silent, the WCAG run left 34 of 65.



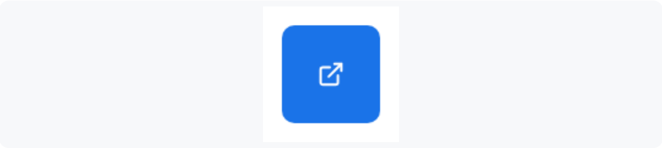
functional_0001 · WCAG prompt

CORRECT ALT

"Email us"

MODEL ALT

"Empty"



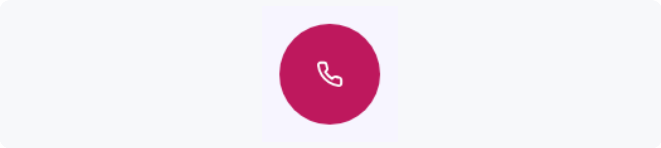
functional_0002 · WCAG prompt

CORRECT ALT

"Open in a new tab"

MODEL ALT

"Empty"



functional_0003 · WCAG prompt

CORRECT ALT

"Call us"

MODEL ALT

"Empty"

CONDITION	CORRECT	HARMLESS	DEGRADED	SILENT	MISLEADING
Naive prompt	29	1	33	0	2
WCAG prompt	13	0	18	34	0
WCAG plus context	31	7	12	2	13

Qwen2-VL-2B, functional images only. Severity is what a correctness score cannot see.

Purpose is in the DOM, not the pixels

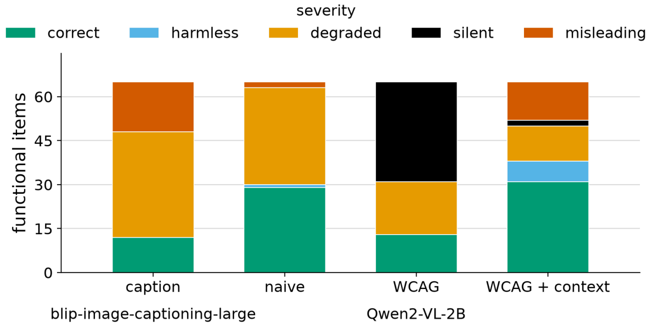
WCAG PROMPT

20% →

WCAG PLUS CONTEXT

58%

Same model, same images, same rules in the prompt. The only addition is the element tag, its HTML, and the text next to it. Functional accuracy goes up 38 points.



Not a free win at this size: with context the model over-reads the page. Accuracy on decorative 100% to 14% and image of text 86% to 34%. The practical version still holds: a generator that sees only the image cannot recover interactive purpose, and anything running in a browser already has the DOM.

LIMITATIONS

What this does not show

- Images are generated, not scraped. Clean and unambiguous, so these numbers are an upper bound on real web images.
- 2 models evaluated, one family plus a captioner baseline. This is not a claim about vision models in general.
- Scoring is rule based on per-item keywords. The judgment calls are in the repo, not hidden: partial credit rules, what counts as a synonym, how padding is treated.
- n = 298, so per-category cells are small. The 30 item second rater subset is in the repo and not yet scored, so no agreement number is claimed.

The effect on functional images is large enough to survive all of this. The exact percentages are not the claim, the direction is.

TAKEAWAY

Compliance prompting moves the failure out of sight

A model can apply the decorative rule and the text rule. It cannot infer what a control does, because that is not a visual property.

The visible failure, a description where a verb belonged, becomes an invisible one: nothing announced at all. Correctness scores rate these the same. Users do not.

The fix is not a better prompt. It is giving the model the context that carries function, which anything running in a page already has.

https://github.com/cansirin/ai-mrp-summer-2026-alt-text

Image generator, model runs, scorer, and the summary file every number in this deck is read from.